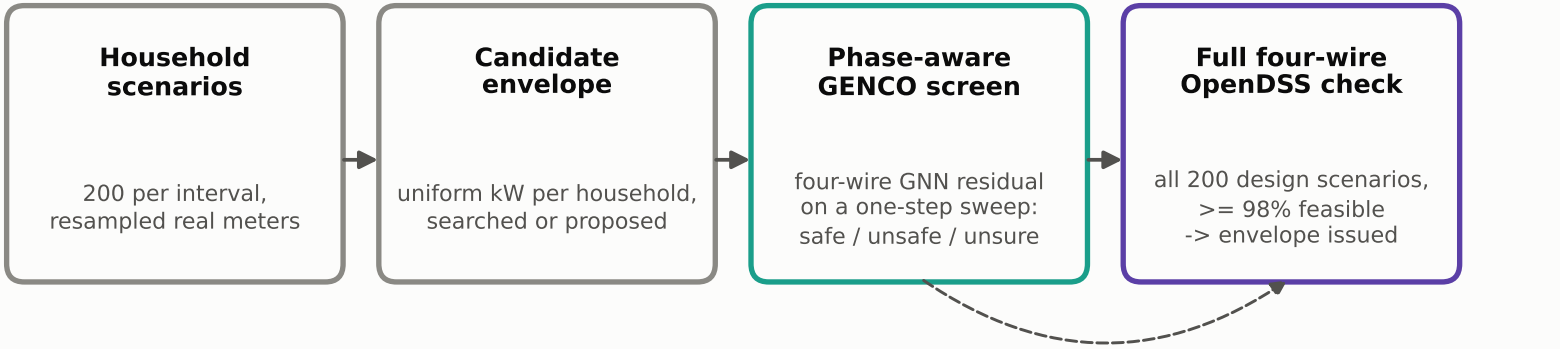
Household scenarios
200 per interval, resampled real meters
Candidate envelope
uniform kW per household, searched or proposed
Phase-aware GENCO screen
four-wire GNN residual on a one-step sweep: safe / unsafe / unsure
Full four-wire OpenDSS check
all 200 design scenarios, >= 98% feasible -> envelope issued
unsure states go to OpenDSS (5.8% with the GNN, 35.8% with the prior alone)